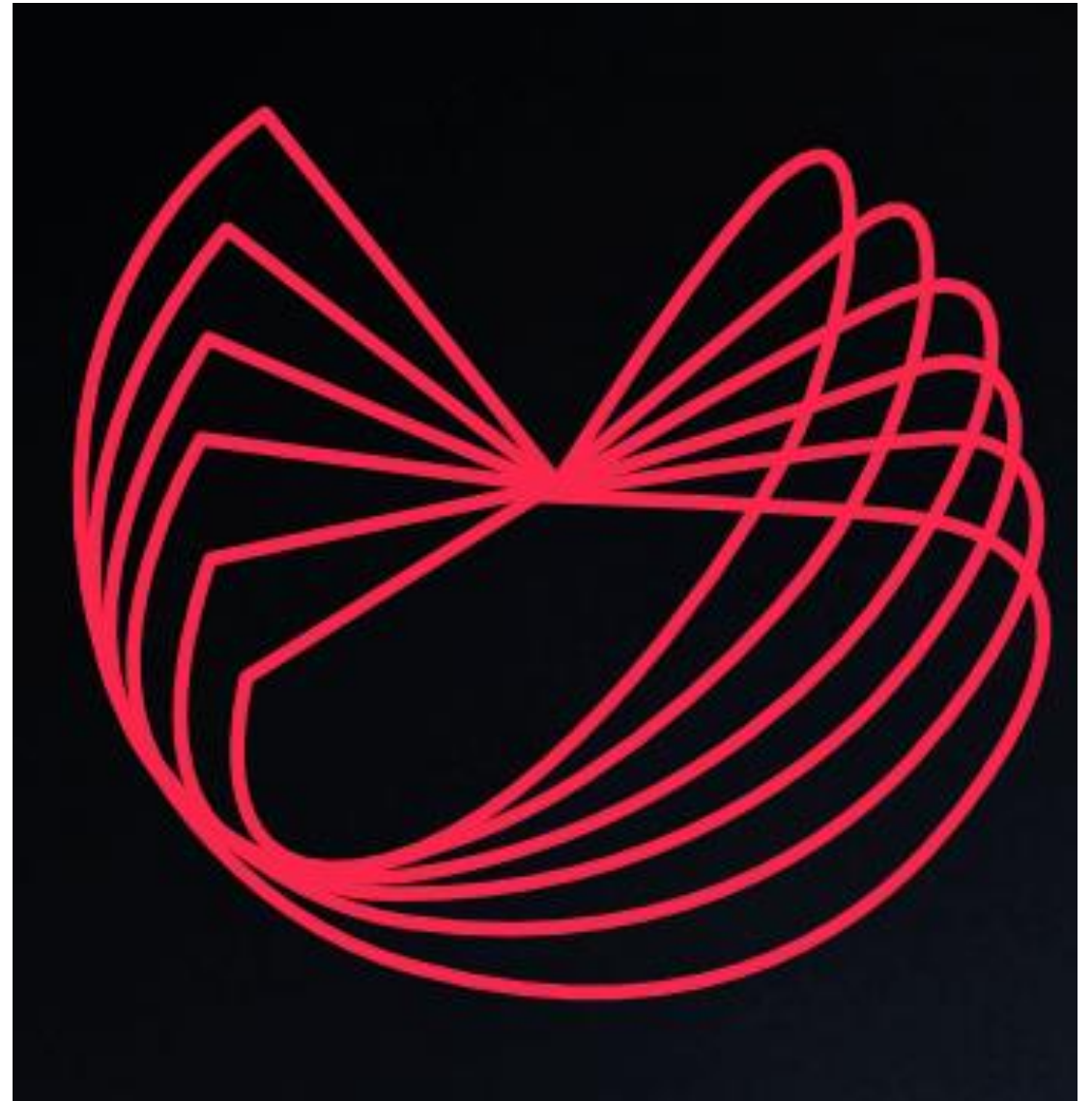

SUMMARY OF THE AUTOPET CHALLENGE SERIES (I – IV)



<https://www.autopet.org>

Organizers



INTRODUCTION

Importance of PET/CT

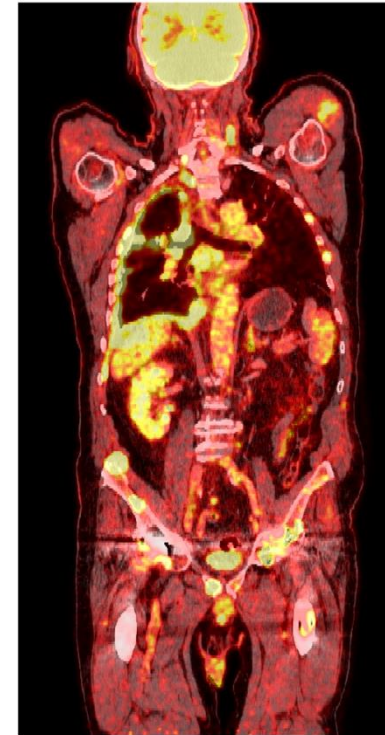
- Combines metabolic (PET) + anatomical (CT) information
- Enables precise tumor localization in oncology

Clinical relevance of PET/CT tumor segmentation

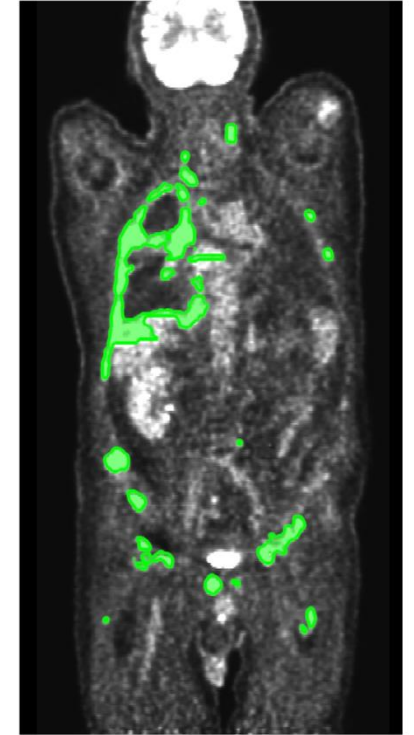
- Supports radiotherapy planning
- Helps monitor treatment response
- Improves disease staging and prognosis

Clinical problem

- Manual contouring is time consuming
- Manual contouring is prone to inter-observer variability



whole-body
FDG-PET/CT
image fusion



manual segmentation
of FDG-avid
malignant lesions

INTRODUCTION

Potential solution

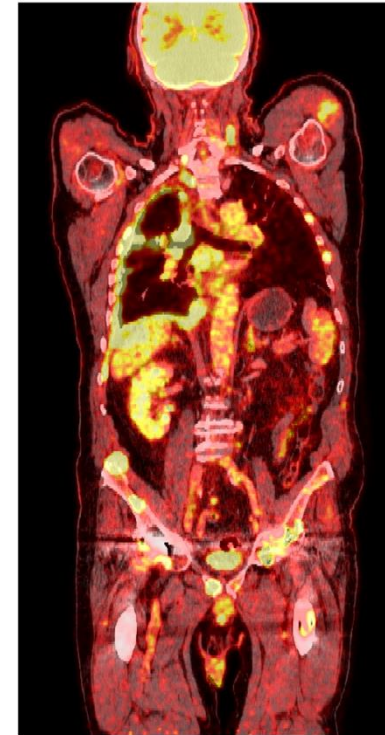
AI-based automated segmentation of lesions in PET/CT

Why automated segmentation is difficult

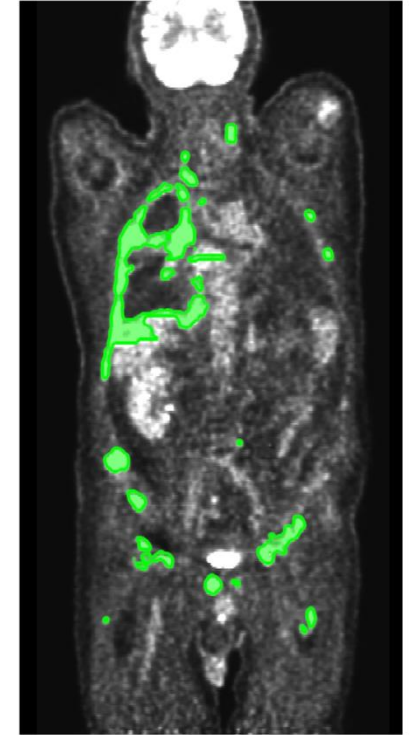
- Limited availability of well curated data cohorts
- Difficulty in selecting the optimal model outputs

Role of the autoPET challenge series

- Standardized benchmark for PET/CT tumor segmentation
 - Provides large, expert-curated and annotated datasets
 - Enables fair comparison of methods
 - Drives reproducible research and methodological progress
-



whole-body
FDG-PET/CT
image fusion



manual segmentation
of FDG-avid
malignant lesions

OVERVIEW OF THE AUTOPET CHALLENGE SERIES (2022 – 2025)

Dataset

Release the largest whole-body PET/CT datasets with real-world clinical variability

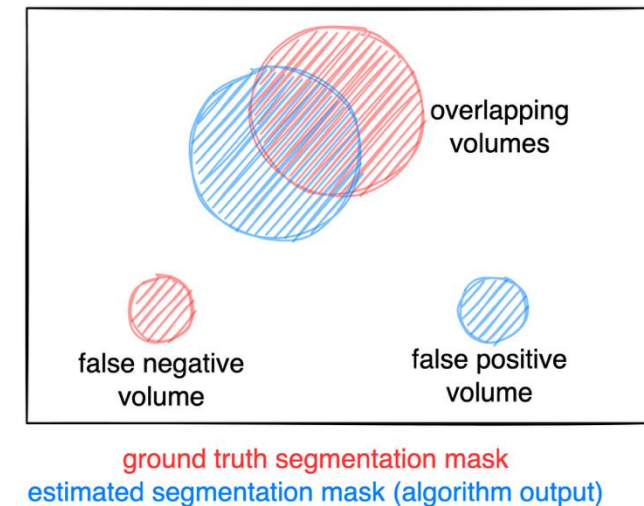
- 1014 FDG¹ PET/CT (900 patients) DOI 10.7937/gkr0-xv29 DOI 10.57754/FDAT.wf9fy-txq84
- 597 PSMA² PET/CT (378 patients) DOI 10.7937/r7ep-3x37 DOI 10.57754/FDAT.6gjsq-zcg93
- 300 longitudinal CT³ (300 patients) DOI 10.57754/FDAT.qwsry-7t837

Main task

Automatic tumor lesion segmentation in whole-body PET/CT

Evaluation metrics

- Dice Score (DSC)
- False Positive Volume (FPV)
- False Negative Volume (FNV)



OVERVIEW OF THE AUTOPET CHALLENGE SERIES (2022 – 2025)

autoPET-I

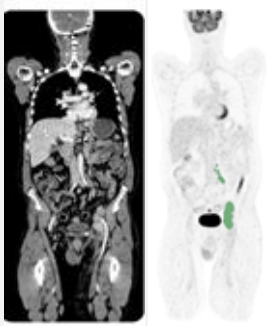
autoPET-II

autoPET-III

autoPET-IV

Segmentation Frontier

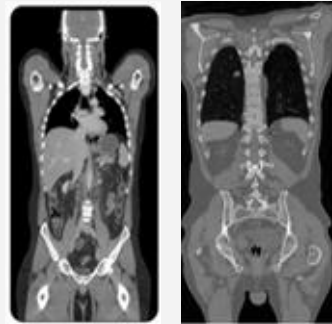
FDG-UKT¹



Generalization Frontier

UKT¹

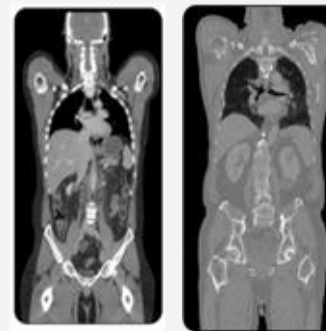
LMU²



Tracer Frontier

FDG¹

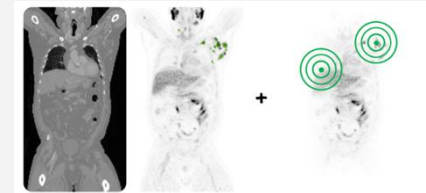
PSMA²



Human Frontier

FDG¹/
PSMA²

Human
clicks



autoPET-I: Segmentation frontier¹

Goal

Automated whole-body FDG-PET/CT lesion segmentation focusing on accuracy, and false positive reduction.

Dataset

- Training cohort (whole-body FDG PET/CT):
 - 1,014 FDG studies at University Hospital Tübingen (UKT)
- Test cohort (whole-body FDG PET/CT) :
 - UKT: 100 studies
 - University Hospital of the LMU in Munich: 50 studies

autoPET-I: Segmentation frontier¹

Architecture and metrics

Training configuration				Evaluation metrics		
Team	Architecture	Post-processing	loss	DSC	FNV[ml]	FPV[ml]
Blackbean	U-Net (3D)	Removal of small cc	DSC + CE	0.6226	0.5445	2.8372
BDAV	U-Net (3D) + U-Net (2D)	Ensembling (2D, 3D)	DSC + CE	0.6208	0.7518	3.6111
nnU-Net (Baseline)	U-Net (3D)	-	DSC + CE	0.662	1.2423	9.3157

Table 1: autoPET-I summary of the baseline and top 2 winning solutions.

cc: connected components, CE: Cross Entropy

autoPET-I: Segmentation frontier¹

Lessons learned

- **Limited generalization:** Models trained on single-institution data showed reduced performance on external datasets (scanner/hospital variability)
- **Small lesion detection remains challenging:** Algorithms frequently missed small or low-uptake lesions, leading to higher false negatives
- **Dataset composition strongly affects performance:** Factors such as training data diversity and patient characteristics influence model robustness

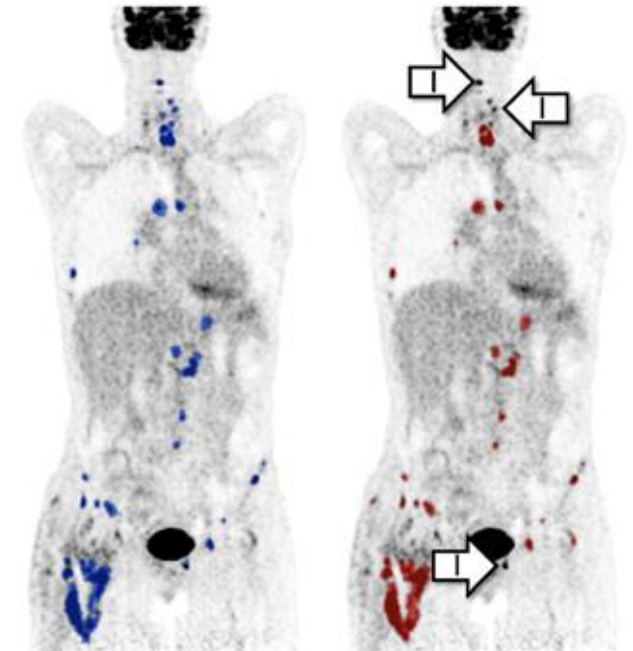


Fig 1: PET image example of study with small average lesion volume. Arrows indicate small lesions that were missed by automated segmentation using the nnU-Net baseline.

autoPET-II: Generalization Frontier⁴

Goal

Automated whole-body PET/CT lesion segmentation focusing on cross-domain robustness and generalization.

Dataset

- **Training cohort** (whole-body FDG PET/CT):
 - 1,014 FDG studies at UKT
- **Test cohort** (whole-body PET/CT):
 - UKT: 50 FDG studies
 - UKT-Patho: 25 FDG studies
 - LMU (Munich): 25 FDG studies
 - LMU-PSMA: 50 PSMA studies
 - UKE-Pediatric (Essen): 50 FDG studies

Award categories

- **Category 1:** Highest overall score
 - **Category 2:** Highest robustness across domains
 - **Category 3:** Jury prizes for scientific contribution and creative engineering
-

autoPET-II: Generalization frontier⁴

Architecture and metrics

Training configuration				Evaluation metrics		
Team	Architecture	Post-processing	loss	DSC	FNV[ml]	FPV[ml]
BAMF Health	U-Net (3D)	Remove additional labels and keep only lesion	DSC + CE	0.5038	8.4154	87.839
Blackbean	U-Net (3D)	-	DSC + CE	0.5198	9.7205	161.392
nnU-Net (Baseline)	U-Net (3D)	-	DSC + CE	0.5024	18.9489	176.43

Table 2: autoPET-II-Category 1 summary of the baseline and top 2 winning solutions.

CE: Cross Entropy, Category 1: Highest overall score

autoPET-II: Generalization frontier⁴

Lessons learned

- **Domain generalization remains difficult and highly domain-dependent:** Performance varied widely across datasets, with notable degradation on out-of-domain data, e.g. pediatric FDG and PSMA datasets (Fig 2).
- **False positives from physiological uptake are the main failure mode:** Physiological uptake often misclassified as lesions, e.g., bladder, liver, salivary glands, spleen, abdomen (Fig 3).

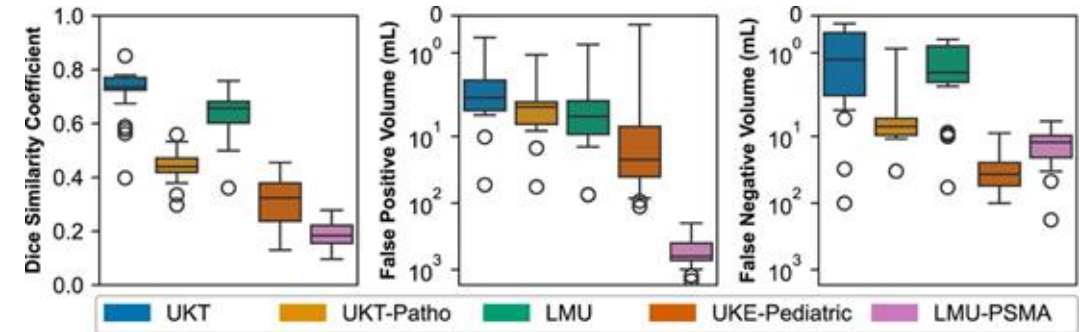


Fig 2: Distribution of mean performances of participating teams.

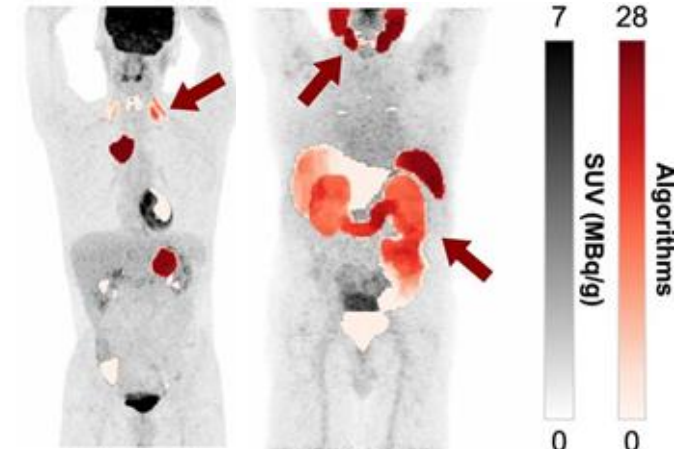


Fig 3: Common observed error patterns in algorithm predictions. Segmentation errors due to physiological uptake in salivary and lacrimal glands, bowel, bladder, liver, spleen, and gallbladder.

autoPET-III: Tracer Frontier

Goal

Automated whole-body PET/CT lesion segmentation focusing on multi-tracer robustness and cross-site generalization.

Dataset

- **Training cohort** (whole-body PET/CT):
 - 1,014 FDG studies at UKT
 - 597 PSMA studies at LMU
- **Test cohort** (whole-body PET/CT):
 - UKT-FDG: 50 studies
 - UKT-PSMA: 50 studies
 - LMU-FDG: 50 studies
 - LMU-PSMA: 50 studies

Award categories

- **Category 1- Best generalizing model:** Develop a model that generalizes well across tracers (FDG, PSMA), optionally using multiple models and additional public data.
 - **Category 2- Datacentrism:** Improve performance by only enhancing the dataset itself (e.g., through data cleaning, outlier handling, augmentation, pre-/post-processing, etc).
-

autoPET-III: Tracer frontier

Architecture and metrics

Training configuration				Evaluation metrics		
Team	Architecture	Post-processing	loss	DSC	FNV[ml]	FPV[ml]
LesionLocator	U-Net (3D)		DSC + CE	0.6616	3.1842	2.7849
IKIM	U-Net (3D)		DSC + CE	0.6496	3.2451	3.0438
nnU-Net (Baseline)	U-Net (3D)	-	DSC + CE	0.4523	7.3355	31.5028

Table 3: autoPET-III-Category 1 summary of the baseline and top 2 winning solutions.

CE: Cross Entropy, Category 1: Best generalizing model

autoPET-III: Tracer frontier

Lessons learned

- **Releasing the new PSMA data improved segmentation performance:** This showed the value of expanding tracer diversity in the dataset.
- **Systematic errors due to physiological uptake persist (Fig 4)**
- **Detection performance depends strongly on lesion size and SUV uptake:** Small and low-uptake lesions remain difficult to segment
- **Multi-tracer modeling showed limited benefits:** Global models and tracer-specific models achieved very similar performance
- **Data-centric approaches remain underexplored:** Potential, but only few participations in this track

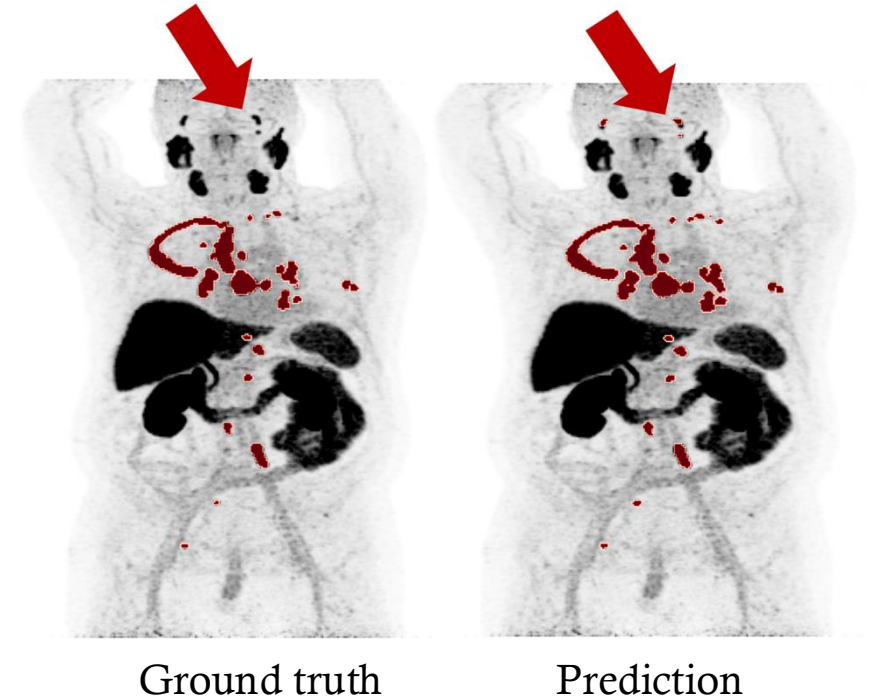


Fig 4: Common observed error patterns in algorithm predictions. Segmentation errors physiological uptake in lacrimal glands (arrow).

autoPET/CT-IV: Human Frontier

Goal

Automated lesion segmentation focusing on robustness under varying levels of simulated human annotations in single-timepoint PET/CT and longitudinal CT.

Dataset

- **Training cohort** (whole-body PET/CT):
 - 1,014 FDG studies at UKT
 - 597 PSMA studies at LMU
 - Simulated human annotations (foreground/background clicks)
- **Test cohort** (whole-body PET/CT):
 - UKT-FDG: 50 studies
 - UKT-PSMA: 50 studies
 - LMU-FDG: 50 studies
 - LMU-PSMA: 50 studies
 - Simulated human annotations (clicks)

Award categories

- **Task 1 - Single-staging whole-body PET/CT:** Investigate the impact of simulated human clicks in a multi-tracer and multi-center imaging setting.
 - **Task 2 - Lesion Segmentation in Longitudinal CT:** Investigate how to integrate information (CT + center clicks in the lesion region) from baseline scans to accurately segment the lesion in a follow-up scan.
-

autoPET/CT-IV: Human frontier

Architecture and metrics

Training configuration				Evaluation metrics		
Team	Architecture	Post-processing	loss	DSC	FNV[ml]	FPV[ml]
BIRTH	U-Net (3D)	PET SUV thresholding	DSC + CE	0.7412	2.1665	1.2898
LesionLocator	U-Net (3D)	-	DSC + CE	0.7236	1.7511	2.4750
nnU-Net (Baseline)	U-Net (3D)	-	DSC + CE	0.6349	3.0666	10.8394

Table 4: autoPET-IV-Task 1 summary of the baseline and top 2 winning solutions.

CE: Cross Entropy, Task 1: Single-staging whole-body PET/CT

autoPET/CT-IV: Human frontier

Lessons learned

- The human-in-the-loop setting with simulated clicks consistently reduced false negatives and improved segmentation accuracy (Fig 5, 6).
- Increasing the number of clicks also increased false positives, highlighting a sensitivity-specificity trade-off (Fig 6).
- Regions with low contrast and high physiological uptake remain challenging for accurate segmentation.
- Segmenting evolving lesions from baseline to follow-up proved more complex and underexplored than static segmentation: This is reflected by substantially lower performance of the winning solution compared to Task 1 and fewer participating teams.

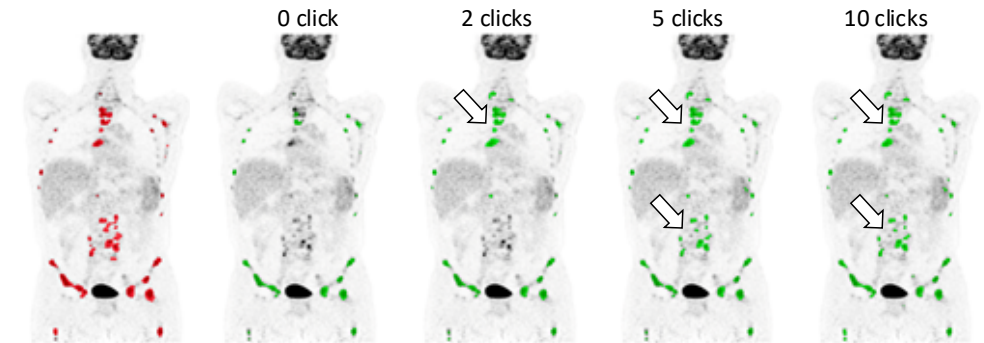


Fig 5: PET image with manual segmentation (red) and corresponding nnU-Net baseline predicted masks (green) at different interaction levels.

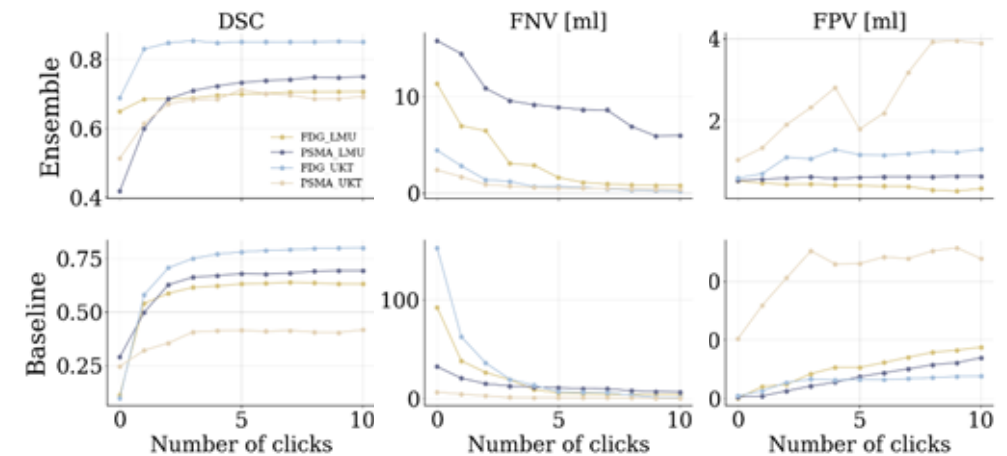
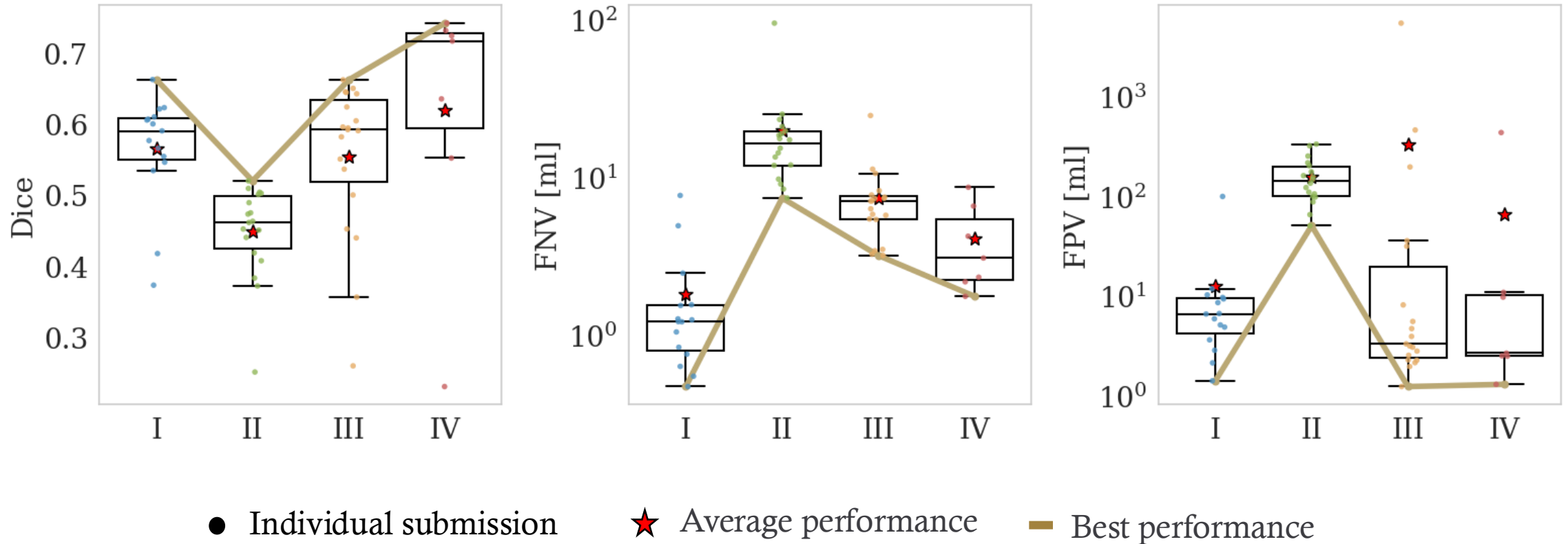


Fig 6: Metrics as a function of the number of clicks across tracers and center.

autoPET I-IV Performance Summary



CONCLUSION

- autoPET I-IV demonstrate the maturation of PET/CT segmentation from proof-of-concept to clinical orientation.
- This progression is accompanied by a general improvement in segmentation performance.
- **autoPET-V is waiting for you. Please visit the official challenge website via the link below and join us in advancing robust and clinically meaningful medical image analysis**

<https://autopet-v.grand-challenge.org/>



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